



JKBOSE 2026

CLASS 11TH



SEQUENCES AND SERIES

MATHS

LECTURE-05

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TOPICS TO BE COVERED

1

Exercise Discussion 8.2

Part 2

2

Related Problems

$$\frac{a}{b} = \frac{c}{d}$$

$$\frac{a+b}{a-b} = \frac{c+d}{c-d} \quad (\text{componendo-dividendo}).$$

EXERCISE – 8.2

QUESTION

How many terms of G.P. $3, 3^2, 3^3, \dots$ Are needed to give the sum 120?

(Sol) let $S_n = 120$

$$\Rightarrow \frac{a(r^n - 1)}{r - 1} = 120$$

$$\Rightarrow \frac{3(3^n - 1)}{3 - 1} = 120$$

$$\Rightarrow \frac{3(3^n - 1)}{2} = 120$$

$$\Rightarrow 3^n - 1 = \frac{120 \times 2}{3} = 80$$

$$\Rightarrow 3^n = 81$$

$$\Rightarrow 3^n = 3^4$$

$$\Rightarrow n = 4$$

Hence the result.

QUESTION

$$\frac{a, ar, ar^2}{16}, \frac{ar^3, ar^4, ar^5}{128}$$



The sum of first three terms of a G.P. is 16 and the sum of the next three terms is 128.
Determine the first term, the common ratio and the sum to n terms of the G.P.

(Sol) Given $a + ar + ar^2 = 16$

$$\Rightarrow a(1+r+r^2) = 16 \quad \text{--- (1)}$$

Also, $ar^3 + ar^4 + ar^5 = 128$

$$\Rightarrow ar^3(1+r+r^2) = 128 \quad \text{--- (2)}$$

Dividing (2) by (1), we get,

$$\frac{\cancel{ar^3}(1+\cancel{r}+\cancel{r^2})}{\cancel{a}(1+\cancel{r}+\cancel{r^2})} = \frac{\cancel{128}^8}{\cancel{16}_1} = 8$$

$$\Rightarrow r^3 = 8 = 2^3$$

$$\Rightarrow r = 2$$

so from (1), we get,

$$a(1+2+2^2) = 16$$

$$\Rightarrow a(7) = 16$$

$$\Rightarrow a = \frac{16}{7}$$

Therefore $S_n = \frac{a(r^n - 1)}{r - 1}$

$$\Rightarrow S_n = \frac{16}{7}(2^n - 1)$$

Hence the result.

QUESTION

Give a G.P. with $a = 729$ and 7th term 64, determine S_7 .

H/W

QUESTION



Find a G.P. for which sum of the first two terms is -4 and the fifth term is 4 times the third term.

(Sol) Given $a + ar = -4$

$$\Rightarrow a(1+r) = -4 \quad \text{--- (1)}$$

Also, $a_5 = 4a_3$

$$\Rightarrow a r^4 = 4 \times a r^2$$

$$\Rightarrow r^2 \times r^2 = 4 r^2$$

$$\Rightarrow r^2 = 4$$

$$\Rightarrow r = \pm 2$$

So from (1), we get;

$$a(1+2) = -4, \quad a(1-2) = -4.$$

$$\Rightarrow a = -\frac{4}{3}, 4.$$

Therefore the required G.P.
 $a, ar, ar^2, ar^3, \dots$

$$\Rightarrow -\frac{4}{3}, -\frac{8}{3}, -\frac{16}{3}, \dots$$

$$\text{or } 4, -8, 16, -32, \dots \text{ Hence the result.}$$

QUESTION

If the 4th, 10th and 16th terms of a G.P. are x , y and z , respectively. Prove that x , y , z are in G.P.

$$a_4 = x$$

$$a_{10} = y$$

$$a_{16} = z$$

$$ar^3 = x, ar^9 = y, ar^{15} = z$$

$$\Rightarrow \frac{y}{x} = r^6, \frac{z}{y} = r^6$$

$$\Rightarrow \frac{y}{x} = \frac{z}{y}$$

$$\Rightarrow x, y, z \text{ are in G.P.}$$

$$a, ar, ar^2, ar^3, ar^4, ar^5 \dots$$

$$ar, ar^3, ar^5, \dots$$

QUESTION

Find the sum of the products of the corresponding terms of the sequences 2, 4, 8, 16, 32
and 128, 32, 8, 2, $\frac{1}{2}$.

H/w

QUESTION

Show that the products of the corresponding terms of the sequences $a, ar, ar^2, \dots, ar^{n-1}$ and $A, AR, AR^2, \dots, AR^{n-1}$ form a G.P, and find the common ratio.

HP/W

QUESTION

$$a, ar, ar^2, ar^3$$



Find four numbers forming a geometric progression in which the third term is greater than the first term by 9, and the second term is greater than the 4th by 18.

(Sol) Given that: $ar^2 = a + 9$
 $\Rightarrow a(r^2 - 1) = 9$ — (1)

Also, $ar = ar^3 + 18$.

$$\Rightarrow ar(1 - r^2) = 18$$

$$\Rightarrow ar(r^2 - 1) = -18$$
 — (2)

Dividing (2) by (1), we get:

$$\frac{ar(r^2 - 1)}{a(r^2 - 1)} = \frac{-18}{9}$$

$$\Rightarrow r = -2$$

So, from (1), we get;

$$a(4 - 1) = 9$$

$$\Rightarrow a = 3$$

Therefore the required four numbers are
3, -6, 12, -24

QUESTION

If the p^{th} , q^{th} and r^{th} terms of a G.P. are a , b and c , respectively. Prove that

$$a^{q-r} b^{r-p} c^{p-q} = 1. \quad \text{H/W}$$

QUESTION

If the first and the n^{th} term of a G.P. are a and b , respectively, and if P is the product of n terms, prove that $P^2 = (ab)^n$.

H/W.

QUESTION

$$a, ar, ar^2, ar^3, \dots, ar^{n-1}, \underbrace{ar^n, ar^{n+1}, \dots, ar^{2n-1}}_{\text{S'}}$$



Show that the ratio of the sum of first n terms of a G.P. to the sum of terms from $(n+1)^{\text{th}}$ to $(2n)^{\text{th}}$ term is $\frac{1}{r^n}$.

Proof: let $S_n = a + ar + ar^2 + \dots + ar^{n-1}$

$$\Rightarrow S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{--- (1)}$$

Also, let $S' = ar^n + ar^{n+1} + \dots + ar^{2n-1}$

$$\Rightarrow S' = (a + ar + ar^2 + \dots + ar^{n-1} + ar^n + ar^{n+1} + \dots + ar^{2n-1}) - (a + ar + ar^2 + \dots + ar^{n-1})$$

$$\Rightarrow S' = \frac{a(r^{2n} - 1)}{r - 1} - S_n$$

$$\Rightarrow \frac{S'}{S_n} = \frac{a(r^{2n} - 1)}{r - 1} \times \frac{1}{S_n} - 1$$

$$\Rightarrow \frac{S'}{S_n} = \frac{a(r^n - 1)(r^n + 1)}{(r - 1)} \times \frac{(r - 1)}{a(r^n - 1)} - 1 \quad (\text{from (1)})$$

$$\Rightarrow \frac{S'}{S_n} = r^n + 1 - 1 = r^n$$

$$\Rightarrow \frac{S'}{S_n} = \frac{1}{r^n}$$

Hence proved.

QUESTION (PYQ JKBOSE)

$$\left[a^2 r (1+r^2+r^4) \right]^2$$



If a, b, c and d are in G.P. show that $(a^2 + b^2 + c^2)(b^2 + c^2 + d^2) = (ab + bc + cd)^2$.

Proof: let $a = a, b = ar, c = ar^2, d = ar^3$

Consider the following:

$$\begin{aligned} (a^2 + b^2 + c^2)(b^2 + c^2 + d^2) \\ = (a^2 + a^2 r^2 + a^2 r^4)(a^2 r^2 + a^2 r^4 + a^2 r^6) \\ = a^2(1 + r^2 + r^4) \times a^2 r^2(1 + r^2 + r^4) \end{aligned}$$

$$\Rightarrow (a^2 + b^2 + c^2)(b^2 + c^2 + d^2) = a^4 r^2 (1 + r^2 + r^4)^2 \quad \text{--- (1)}$$

$$\text{Also } (ab + bc + cd)^2 = (a^2 r + a^2 r^3 + a^2 r^5)^2$$

$$\Rightarrow (ab + bc + cd)^2 = a^4 r^2 (1 + r^2 + r^4)^2 \quad \text{--- (2)}$$

From (1) & (2), we get,

$$(a^2 + b^2 + c^2)(b^2 + c^2 + d^2) = (ab + bc + cd)^2$$

Hence proved.

QUESTION



Find the value of n so that $\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ may be the geometric mean between a and b .

$$a^{n+\frac{1}{2}+\frac{1}{2}} = a^{n+1} = a^{n+1}$$

$$b^{n+\frac{1}{2}+\frac{1}{2}} = b^{n+1} = b^{n+1}$$

$$(Sol) \text{ let } \frac{a^{n+1} + b^{n+1}}{a^n + b^n} = \sqrt{ab} = a^{\frac{1}{2}} b^{\frac{1}{2}}$$

$$\Rightarrow a^{n+1} + b^{n+1} = a^{\frac{1}{2}} b^{\frac{1}{2}} (a^n + b^n)$$

$$\Rightarrow a^{n+1} + b^{n+1} = a^{\frac{n+1}{2}} b^{\frac{1}{2}} + a^{\frac{1}{2}} b^{\frac{n+1}{2}}$$

$$\Rightarrow a^{n+1} - a^{\frac{n+1}{2}} b^{\frac{1}{2}} = a^{\frac{1}{2}} b^{\frac{n+1}{2}} - b^{n+1}$$

$$\Rightarrow a^{\frac{n+1}{2}} (a^{\frac{1}{2}} - b^{\frac{1}{2}}) = b^{\frac{n+1}{2}} (a^{\frac{1}{2}} - b^{\frac{1}{2}})$$

$(a \neq b)$

$$\Rightarrow a^{\frac{n+1}{2}} = b^{\frac{n+1}{2}}$$

$$\Rightarrow \left(\frac{a}{b}\right)^{\frac{n+1}{2}} = 1$$

$$\Rightarrow \left(\frac{a}{b}\right)^{\frac{n+1}{2}} = \left(\frac{a}{b}\right)^0$$

$$\Rightarrow \frac{n+1}{2} = 0$$

$$\Rightarrow n = -\frac{1}{2}$$

$$a^x = a^y$$

$$\Rightarrow x = y$$

$$a \neq 0, -1, 1$$

Hence the result.

QUESTION



The sum of two numbers is 6 times their geometric mean, show that numbers are in the ratio $(3 + 2\sqrt{2}) : (3 - 2\sqrt{2})$.

Proof: let a & b be the required positive numbers, then

$$a + b = 6\sqrt{ab}$$

$$\Rightarrow \frac{a+b}{2\sqrt{ab}} = 3$$

$$\Rightarrow \frac{\sqrt{a}^2 + \sqrt{b}^2}{2\sqrt{ab}} = \frac{3}{1}$$

Applying Componendo-dividendo, we get.

$$\frac{\sqrt{a}^2 + \sqrt{b}^2 + 2\sqrt{ab}}{\sqrt{a}^2 + \sqrt{b}^2 - 2\sqrt{ab}} = \frac{3+1}{3-1} = \frac{4}{2} = 2.$$

$$\Rightarrow \frac{(\sqrt{a} + \sqrt{b})^2}{(\sqrt{a} - \sqrt{b})^2} = 2 \Rightarrow \left(\frac{\sqrt{a} + \sqrt{b}}{\sqrt{a} - \sqrt{b}} \right)^2 = 2.$$

$$\Rightarrow \frac{\sqrt{a} + \sqrt{b}}{\sqrt{a} - \sqrt{b}} = \frac{\sqrt{2}}{1}$$

Again applying Componendo-dividendo, we get;

$$\frac{\cancel{\sqrt{a}} + \cancel{\sqrt{b}} + \cancel{\sqrt{a}} - \cancel{\sqrt{b}}}{\cancel{\sqrt{a}} + \cancel{\sqrt{b}} - \cancel{\sqrt{a}} + \cancel{\sqrt{b}}} = \frac{\sqrt{2} + 1}{\sqrt{2} - 1}$$

$$\Rightarrow \frac{\cancel{2}\sqrt{a}}{\cancel{2}\sqrt{b}} = \frac{\sqrt{2}+1}{\sqrt{2}-1}$$

$$\Rightarrow \left(\frac{\cancel{\sqrt{a}}}{\sqrt{b}} \right)^2 = \left(\frac{\sqrt{2}+1}{\sqrt{2}-1} \right)^2$$

$$\Rightarrow \frac{a}{b} = \frac{(\sqrt{2}+1)^2}{(\sqrt{2}-1)^2} = \frac{\cancel{\sqrt{2}^2}+1+2\sqrt{2}}{\sqrt{2}^2+1-2\sqrt{2}}$$

$$\Rightarrow \boxed{\frac{a}{b} = \frac{3+2\sqrt{2}}{3-2\sqrt{2}}}$$

Hence proved

QUESTION

If A and G be A.M. and G.M., respectively between two positive numbers, prove that the numbers are $A \pm \sqrt{(A+G)(A-G)}$

Proof: let the required numbers be a & b

$$\text{So, } A = \frac{a+b}{2} \text{ \& } G = \sqrt{ab}$$

$$\Rightarrow a+b=2A, \quad ab=G^2$$

Now, Consider the following:

$$x^2 - (a+b)x + ab = 0$$

$$\Rightarrow x^2 - 2Ax + G^2 = 0$$

$$\Rightarrow x = \frac{2A \pm \sqrt{4A^2 - 4G^2}}{2}$$

$$\Rightarrow x = \frac{2A \pm 2\sqrt{A^2 - G^2}}{2}$$

$$\Rightarrow x = A \pm \sqrt{(A+G)(A-G)}$$

QUESTION

The number of bacteria in a certain culture doubles every hour. If there were 30 bacteria present in the culture originally, how many bacteria will be present at the end of 2nd hour, 4th hour and n th hour?

P/W:

QUESTION

What will Rs. 500 amounts to in 10 years after its deposit in a bank which pays annual interest rate of 10% compounded annually?

h/w

QUESTION

If A.M. and G.M. of roots of a quadratic equation are 8 and 5, respectively, then obtain the quadratic equation. a & b

$$x^2 - (a+b)x + ab = 0$$

$$\Rightarrow x^2 - 16x + 25 = 0.$$

$$\frac{a+b}{2} = 8$$

$$\sqrt{ab} = 5$$

$$a+b = 16$$

$$ab = 25.$$

QUESTION



$$\frac{\frac{5}{16}}{-\frac{5}{4}} = \frac{\cancel{5}}{16} \times \frac{4}{\cancel{-5}} = -\frac{1}{4}$$
$$\delta = -\frac{1}{4}$$

The sum to infinity of the GP $-\frac{5}{4}, \frac{5}{16}, -\frac{5}{64}, \dots$ is :

1 $\times$

$$S_{\infty} = \frac{a}{1-\delta} \text{ , } |\delta| < 1.$$

2 $\times$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

3 -1 ✓

$$S_{\infty} = \frac{\left(-\frac{5}{4}\right)}{\left(1 + \frac{1}{4}\right)} = \frac{-\frac{5}{4}}{\left(\frac{5}{4}\right)} = \frac{\cancel{-5}}{4} \times \frac{4}{\cancel{5}} = -1$$

4 None of these $\times$

$$\frac{\left(-\frac{5}{4}\right)}{1 - \left(-\frac{1}{4}\right)} = \frac{-\frac{5}{4}}{1 + \frac{1}{4}} = \frac{\left(-\frac{5}{4}\right)}{\left(\frac{5}{4}\right)} = -1$$

QUESTION

If a, b, c, d and p are distinct real numbers such that

$$(a^2 + b^2 + c^2)p^2 - 2p(ab + bc + cd) + (b^2 + c^2 + d^2) \leq 0, \text{ then } a, b, c, d \text{ are in}$$

1

A. P.

2

G. P.

3

H. P.

4

None of these

HP/W

Some Special Series

- Sum of first n natural numbers, $\sum n = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$
- Sum of the squares of first n natural numbers, $\sum n^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$
- Sum of the cubes of first n natural numbers, $\sum n^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$

H/W

QUESTION

Sum up to 16 terms of the series $\frac{1^3}{1} + \frac{1^3+2^3}{1+3} + \frac{1^3+2^3+3^3}{1+3+5} + \dots$ is :

H/W

1 450

2 456

3 446

4 None of these

Thank
You